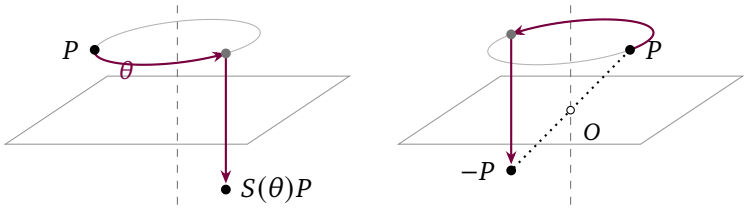
# Symmetries in Physics — Exercise Sheet 7

*Molecular vibrations*

**Variant: Questions only**

**Exercise 1.** The table in Lecture 7 lists $\chi_{\text{vec}}$ for the rotations $C_2, C_3, C_4, C_6$, the mirror plane $\sigma = S(0)$ and the inversion $i = S(\pi)$. It omits the rotoreflections $S_n = S(2\pi/n)$. A rotoreflection is a rotation through $2\pi/n$ followed by the reflection in the plane perpendicular to its axis (left picture).



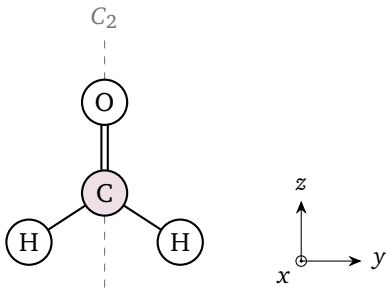

(a) $S(\theta)$ (b) $S_2$

(i) Compute $\chi_{\text{vec}}(S_n)$ for $n = 3, 4, 6$.

(ii) Show that $S_2$ is the inversion (right picture).

(iii) The lemma of the lecture says that a rotoreflection through a nonzero angle leaves only the centre in place. Prove this for $S_n$ with $n \geq 3$. Which values can $N_g$ then take?

**Exercise 2.** Formaldehyde $H_2CO$ is planar. The C–O bond bisects the H–C–H angle. Place it as water was placed in Lecture 7: in the $yz$-plane, with the C–O bond along the $z$ axis. The molecular plane is the page.




(i) Find the point group with the three questions of Lecture 7: the principal axis, the vertical planes, a horizontal plane. Name the planes $\sigma_v(yz)$ and $\sigma'_v(xz)$, as for water.

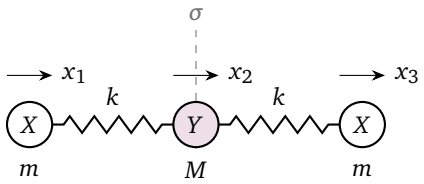
(ii) Using the lemma on atoms left in place, compute $\chi_{\text{disp}}$ and $\chi_{\text{vib}}$.

(iii) Decompose $\Gamma_{\text{vib}}$ with the $C_{2v}$ character table of Lecture 7's water tutorial. With the planes named this way, it reads

| $C_{2v}$ | $E$ | $C_2$ | $\sigma_v(yz)$ | $\sigma'_v(xz)$ | |
|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | $z$ |
| $A_2$ | 1 | 1 | −1 | −1 | |
| $B_1$ | 1 | −1 | 1 | −1 | $y$ |
| $B_2$ | 1 | −1 | −1 | 1 | $x$ |

(iv) A nonlinear planar molecule has $2N-3$ in-plane and $N-3$ out-of-plane vibrations. Check your answer against these counts, using the characters of the molecular plane $\sigma_v(yz)$.

**Exercise 3.** A linear triatomic molecule $X$–$Y$–$X$ has masses $m, M, m$ and springs of stiffness $k$ between neighbours. The atoms move along the molecular axis only, and their displacements $x_1, x_2, x_3$ are all measured in the same direction. The mirror $\sigma$ through $Y$, perpendicular to the axis, exchanges the end atoms. The point group of a linear molecule is infinite, so, as in Lecture 7, we do not use the whole group. We use only the group $\{E, \sigma\}$.

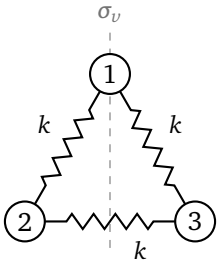



(i) Using the displacement representation of Lecture 7, show that $\rho(\sigma)(x_1, x_2, x_3) = (-x_3, -x_2, -x_1)$, and that $\rho(\sigma)$ commutes with the equations of motion.

(ii) Split $\mathbb{R}^3$ into the even and the odd subspace of $\rho(\sigma)$, block-diagonalise the equations of motion, and find the vibration frequencies.

(iii) Which mode is the symmetric stretch, and which the antisymmetric stretch? The plain exchange $P\colon (x_1, x_2, x_3) \mapsto (x_3, x_2, x_1)$ also commutes with the dynamics. Why does it give the opposite labels?

**Exercise 4.** Three equal masses $m$ sit at the corners of an equilateral triangle, and each pair is joined by a spring of stiffness $k$. The springs are unstretched at equilibrium. The masses move only in the plane of the triangle. The symmetry group is $C_{3v} \cong S_3$: the rotations through $2\pi/3$ and the three mirror lines through the corners, as in Lecture 5.




(i) Adapt the lemma on atoms left in place to the plane: replace $\chi_{\text{vec}}$ by the trace of the $2 \times 2$ matrix $R(g)$ of each operation. Show that $\chi_{\text{disp}} = (6, 0, 0)$ on the classes $(E,\ 2C_3,\ 3\sigma_v)$, and that $\Gamma_{\text{disp}} = A_1 \oplus A_2 \oplus 2E$.

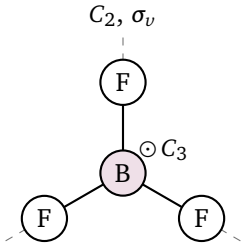
(ii) The two translations transform as $(x, y)$, that is, as $E$. The rotation in the plane is labelled by an angle, which the operation $g$ multiplies by $\det R(g)$. Show that the rotation transforms as $A_2$, and deduce $\Gamma_{\text{vib}} = A_1 \oplus E$.

(iii) The type $A_1$ occurs once, so its symmetry-adapted vector is a normal mode. In it every mass moves radially outward by the same distance $s$ (the *breathing mode*). Show that each spring lengthens by $\sqrt{3}\,s$, and find $\omega^2$.

(iv) The type $E$ occurs twice in $\Gamma_{\text{disp}}$, so by Lecture 6's block theorem the dynamical matrix $D = K/m$ acts on the isotypic component of $E$ through a $2 \times 2$ reduced matrix $F^{(E)}$.

One of its eigenvalues is 0, because the translations lie in this component. Show that $\operatorname{tr} D = \sum_i n_i \operatorname{tr} F^{(i)}$, and use it to find the other eigenvalue, the squared frequency $\omega_E^2$ of the $E$ doublet, without computing $F^{(E)}$.

**Exercise 5 (boron trifluoride, end to end).** Run the full vibrational analysis of planar $BF_3$. Its point group is $D_{3h}$. A B atom sits at the centre, and three F atoms sit at the corners of an equilateral triangle. The page is the molecular plane $\sigma_h$. The $C_3$ axis is perpendicular to it through B. Each $C_2$ axis runs along a B–F bond. Each mirror $\sigma_v$ contains the $C_3$ axis and one $C_2$ axis.




(i) Write down $\chi_{\text{disp}}$ from the lemma on atoms left in place (Lecture 7).

(ii) Reduce it, subtract translations and rotations, and obtain $\Gamma_{\text{vib}}$.

(iii) Two types occur once in $\Gamma_{\text{vib}}$, so their modes are fixed by symmetry alone. Draw them.

*Data.* $D_{3h}$ has classes $E$, $2C_3$, $3C_2$, $\sigma_h$, $2S_3$, $3\sigma_v$, of sizes $1, 2, 3, 1, 2, 3$. Here the *operation* $S_3$ (not the group) is the rotoreflection: rotation by $2\pi/3$ followed by reflection in $\sigma_h$. The complete character table is

| $D_{3h}$ | $E$ | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|---|---|---|---|---|---|---|
| $A_1'$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $A_2'$ | 1 | 1 | −1 | 1 | 1 | −1 |
| $E'$ | 2 | −1 | 0 | 2 | −1 | 0 |
| $A_1''$ | 1 | 1 | 1 | −1 | −1 | −1 |
| $A_2''$ | 1 | 1 | −1 | −1 | −1 | 1 |
| $E''$ | 2 | −1 | 0 | −2 | 1 | 0 |

The class sizes are the coefficients in the headings. One prime marks a type even under $\sigma_h$, and two primes a type odd under it. Translations transform as $z \sim A_2''$, $(x, y) \sim E'$, and rotations as $R_z \sim A_2'$, $(R_x, R_y) \sim E''$.